

# Didactic Manual

**big.bi  $\rightarrow$  bi.gbi**

**Articulatory Simulation of the Verbal Transformation**

Reproduction of the demonstrations described in

*“Why can big.bi be changed to bi.gbi?”*

*A mathematical model of syllabification and articulatory synthesis”*

arXiv:2307.02299 (Berthommier, July 2023)

**Model:** VLAM (Maeda, 7 parameters)

**Code:** Python 3.9+ (MIT License)

**Dependencies:** numpy, scipy, matplotlib, opencv-python

**Repository:** bigbi-article-repo

*This manual provides step-by-step instructions for reproducing the verbal transformation simulations, including the polar trajectory visualization, the U-shaped Tp variation, and the comparison with the article’s Figure 4.*

# Contents

|          |                                                                       |          |
|----------|-----------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                   | <b>2</b> |
| 1.1      | What this repository provides . . . . .                               | 2        |
| 1.2      | The VLAM model . . . . .                                              | 2        |
| 1.3      | Article parameters . . . . .                                          | 3        |
| <b>2</b> | <b>The Mathematical Model</b>                                         | <b>4</b> |
| 2.1      | Polar coordinates and targets . . . . .                               | 4        |
| 2.2      | Arc trajectories . . . . .                                            | 4        |
| 2.3      | Anchoring vowels: Vo and Ve . . . . .                                 | 4        |
| 2.4      | Superposition . . . . .                                               | 4        |
| <b>3</b> | <b>The Syllable Graph Architecture</b>                                | <b>5</b> |
| 3.1      | CV graph . . . . .                                                    | 5        |
| 3.2      | CVC graph . . . . .                                                   | 5        |
| 3.3      | CCV graph (clusters) . . . . .                                        | 5        |
| 3.4      | The fusion mechanism . . . . .                                        | 5        |
| <b>4</b> | <b>The <math>\delta_o</math> and <math>\delta_e</math> Parameters</b> | <b>6</b> |
| 4.1      | Implementation . . . . .                                              | 6        |
| 4.2      | Effect on the anchoring vowels . . . . .                              | 6        |
| 4.3      | The Ve plateau fix . . . . .                                          | 6        |
| <b>5</b> | <b>Reproduction Guide</b>                                             | <b>7</b> |
| 5.1      | Installation . . . . .                                                | 7        |
| 5.2      | Demonstration 1: big.bi → bi.gbi (Article §4) . . . . .               | 7        |
| 5.3      | Demonstration 2: U-shaped $T_p$ variation . . . . .                   | 7        |
| 5.4      | Demonstration 3: Comparison with the article . . . . .                | 7        |
| <b>6</b> | <b>Polar Trajectory Reconstruction</b>                                | <b>8</b> |
| 6.1      | Reconstruction rules . . . . .                                        | 8        |
| 6.2      | Display curvature . . . . .                                           | 8        |
| 6.3      | Comparison with the article’s Figure 4 . . . . .                      | 8        |
| <b>7</b> | <b>Results and Duration Analysis</b>                                  | <b>9</b> |
| 7.1      | Duration reduction . . . . .                                          | 9        |
| 7.2      | Perceptual salience of the schwa @ . . . . .                          | 9        |
| <b>8</b> | <b>Code Structure</b>                                                 | <b>9</b> |

|           |                                                   |           |
|-----------|---------------------------------------------------|-----------|
| 8.1       | Modified files (vs original timit-to-Maeda)       | 9         |
| 8.2       | Simulation scripts                                | 10        |
| <b>9</b>  | <b>Quick Demo</b>                                 | <b>10</b> |
| <b>10</b> | <b>Article Demonstrations</b>                     | <b>11</b> |
| 10.1      | Demo 1: /ibia/ VCV synthesis (Figure 1)           | 11        |
| 10.2      | Demo 2: Locus equations (Figure 3)                | 11        |
| 10.2.1    | Root cause of the previous /b-/d/ inversion       | 12        |
| 10.3      | Demo 4: Trough effect (Figure 1b)                 | 13        |
| 10.4      | Demo 5: Six consonant cluster pairs               | 14        |
| 10.5      | Original article simulations (exact reproduction) | 14        |
| 10.5.1    | Figure 1: /ibia/ with T=100ms                     | 14        |
| 10.5.2    | Figure 4: big.bi ( $\delta = 1$ ) and bi.gbi      | 14        |
| 10.5.3    | Supplement: /ib/ → /bi/                           | 14        |
| <b>11</b> | <b>Interactive HTML Simulations</b>               | <b>15</b> |
| 11.1      | big.bi → bi.gbi (T=16, starts at COEFCEN=0.5)     | 15        |
| 11.2      | ib ib → bi bi (NON-REVERSIBLE U-shape)            | 15        |
| 11.3      | ib ib → bi bi (U-shape, reversible)               | 15        |
| <b>12</b> | <b>Conclusion</b>                                 | <b>15</b> |

## 1 Introduction

This manual documents the Python implementation of the articulatory speech synthesis model described in arXiv:2307.02299 [1]. The model, based on VLAM (a Maeda articulatory synthesizer with 7 parameters), simulates the **verbal transformation** from **big.bi** to **bi.gbi** — a phenomenon where the syllable boundary shifts as the pause between syllables is shortened and the anchoring vowels ( $V_o$ ,  $V_e$ ) approach the main vowel  $V$ .

The article’s central claim (§4) is:

*“When  $\delta_o = \delta_e$  tends to 1 the graph reorganization can occur because two successive superimposed segments of duration  $2 \times 2T$  anchored to the same vowel  $V_o = V_e = V$  can fuse into a single one of duration  $3T$ .”*

This repository provides the code to reproduce this transformation step by step, with audio synthesis, polar trajectory visualization, and accessible video output.

### 1.1 What this repository provides

1. The `synthSYL` package (modified from [FBerthommier/timit-to-Maeda](#)) with the  $\delta_o$  and  $\delta_e$  parameters added to `panphon_pipeline()`.
2. The VLAM synthesizer (`vlam.py`) for audio generation from Maeda parameters.
3. Simulation scripts for three demonstrations:
  - **big.bi** → **bi.gbi** (the article’s main result, §4)
  - **ib ib** → **bi bi** (the classical transformation, §3)
  - U-shaped  $T_p$  variation for perceptual demonstration
4. Polar trajectory reconstruction (two-branch: vocalic  $z_v$  and consonantal  $z_c$ ), using primitives from [FBerthommier/covtl-pipeline](#).
5. Dual-panel video generation (sagittal vocal tract | polar plot) with synchronized audio and phoneme labels.

### 1.2 The VLAM model

VLAM (Vocal tract Learner Articulatory Model) is the Maeda model with 7 articulatory parameters:

| Index | Symbol | Description             |
|-------|--------|-------------------------|
| 0     | J      | Jaw opening             |
| 1     | B      | Tongue body position    |
| 2     | D      | Tongue dorsum height    |
| 3     | T      | Tongue tip position     |
| 4     | LP     | Lip protrusion/rounding |
| 5     | LH     | Vertical lip opening    |
| 6     | Hy     | Larynx height           |

**Indexing convention.** Two numbering conventions coexist and must not be mixed:

- the **article** numbers the parameters 1–7 (1=Jaw, 2=Body, 3=Dorsum, 4=Tip, 5=LipP, 6=LipH, 7=Hy); its selection vectors  $S_c$  and  $S_v$  (e.g.  $S_c = \{1, 2, 6\}$ ) use this 1-based numbering;
- the **code** (and the table above) numbers them 0–6; the registry selection vectors are 0-based, e.g. /b/ has `params=[0,1,5]`.

Correspondence: article  $\{1, 2, 6\} = \text{code } \{0, 1, 5\} = \{\text{Jaw, Body, LipH}\}$ ; article  $\{1, 2, 3, 4\} = \text{code } \{0, 1, 2, 3\} = \{\text{Jaw, Body, Dorsum, Tip}\}$ ; article  $S_v = \{3, 4, 5, 7\} = \text{code } \{2, 3, 4, 6\} = \{\text{Dorsum, Tip, LipP, Hy}\}$ . Throughout this manual,  $S_c/S_v$  sets are quoted in the article’s 1-based numbering (as in the article), while parameter indices mentioned in the text refer to the 0-based code indices unless stated otherwise.

The coordination function (article Eq. 1) maps a polar target  $(\rho, \theta)$  to the 7 Maeda parameters:

$$P_i = \Omega_i + \rho \cdot \Psi_{1,i} \cdot \cos(\Psi_{2,i} - \theta) \quad (1)$$

where  $\Omega$ ,  $\Psi_1$ , and  $\Psi_2$  are calibration constants (article Table 1). The CO matrix in `synthSYL/constants.py` matches the article’s Table 1 almost exactly (only the Tip parameter  $c_0$  differs slightly:  $-2.75$  in `synthSYL` vs  $-3.0$  in the article).

### 1.3 Article parameters

The article (§2.1) specifies the following arc parameters:

- $K_{\text{voy}} = 30$  — vowel arc curvature
- $K = 10$  — consonant arc curvature
- $P_{\text{exp}} = 1$  — velocity profile exponent ( $\rho(t) = \cos(\theta(t)/2)$ , not  $\cos^2$ )
- $T = 16$  — time step (didactic mode)
- $\nu = -1$  — coarticulation index

These are set in `synthSYL/constants.py`:

```
DEFAULT_T: int = 16
DEFAULT_K: int = 10          # consonant arc curvature
DEFAULT_KVOY: int = 30      # vowel arc curvature
DEFAULT_PEXP: int = 1       # velocity profile exponent
DEFAULT_NU: int = -1
```

The article notes: “When  $K$  is large ( $K = 30$  for vowel arcs and  $K = 10$  for consonant arcs), the trajectories become straight lines.” With  $P_{\text{exp}} = 1$ , the velocity profile follows  $\cos(\theta/2)$  (a cosine half-bell) rather than  $\cos^2(\theta/2)$  (a Hanning window).

**Note on display curvature:** The polar trajectory visualization uses  $K = 30$  for both vocalic and consonantal branches (the `covtl`-pipeline reference display curvature), not the engine’s  $K = 10$  /  $K_{\text{voy}} = 30$ . This is a deliberate visualization choice that produces moderate, visible curvature for both branches.

## 2 The Mathematical Model

### 2.1 Polar coordinates and targets

Each phoneme target is a point  $(\rho, \theta)$  in the complex plane:

- **Vowels:**  $\rho \in [0, 1]$ ,  $\theta \in [0, 2\pi)$
- **Consonants:**  $\rho \in [0, 1.2]$ ,  $\theta \in [0, 2\pi)$

For example, the vowel /i/ is at  $(\rho = 0.9, \theta = 5\pi/3)$  and the consonant /b/ is at  $(\rho = 1.2, \theta = \pi/3)$ .

### 2.2 Arc trajectories

Motion between two targets  $(\rho_1, \theta_1)$  and  $(\rho_2, \theta_2)$  is produced by the arc function (article Eq. 2):

$$z(t) = (1 - r_k) \cdot \rho_1 e^{i\theta_1} + r_k \cdot \rho_2 \cdot e^{i(\theta_2 + \frac{\nu}{K} \cdot \text{th})} \quad (2)$$

where  $r_k = \cos^{P_{\text{exp}}}(\text{th}/2)$  is the velocity profile,  $\text{th} \in [0, \pi]$  is a half-turn sweep,  $\nu = \pm 1$  controls the curvature direction, and  $K$  is the curvature constant ( $K = 30$  for display,  $K = 10$  for consonant arcs,  $K_{\text{voy}} = 30$  for vowel arcs in the synthesis engine).

**Important:** The article uses  $P_{\text{exp}} = 1$ , so  $r_k = \cos(\text{th}/2)$  (a cosine half-bell, *without* the power of 2). This is the form used for all demonstrations in this repository. The original **synthSYL** package used  $P_{\text{exp}} = 2$  ( $r_k = \cos^2(\text{th}/2)$ , a Hanning window), which has been corrected to match the article.

### 2.3 Anchoring vowels: Vo and Ve

The article (§2.2) introduces two anchoring vowels:

$$V_o = (\delta_o \cdot \rho_V, \theta_V) \quad (3)$$

$$V_e = (\delta_e \cdot \rho_V, \theta_V) \quad (4)$$

- $V_o$  (onset): anticipation of the vowel before the consonant. The value of  $\delta_o$  determines the degree of anticipation.
- $V_e$  (end): centralization of the coda vowel (schwa-like vocoid). Modeled with  $\delta_e = 0.5$  (salient) or  $\delta_e = 0.7$  (audible).

When  $\delta_o = \delta_e \rightarrow 1$ , both  $V_o$  and  $V_e$  approach  $V$  (the full vowel), enabling the fusion of adjacent syllable segments.

### 2.4 Superposition

The final parameter frame is a blend (article Eq. 4):

$$P(t) - \Omega = \text{Re} [S_v \cdot \Psi \cdot \bar{z}_v(t) + S_c \cdot \Psi \cdot \bar{z}_c(t)] \quad (5)$$

where  $S_v$  and  $S_c$  are selection vectors (binary masks over the 7 parameters),  $z_v(t)$  is the vocalic trajectory, and  $z_c(t)$  is the consonantal trajectory.

### 3 The Syllable Graph Architecture

#### 3.1 CV graph

A CV syllable (e.g., /bi/) has 3 nodes and 4 arcs:

- Nodes:  $V = \{V_o, C, V\}$
- Locations:  $L = \{(\delta_o \rho_V, \theta_V), (\rho_C, \theta_C), (\rho_V, \theta_V)\}$
- Arcs:  $E = \{(V_o, C), (C, V), (V_e, V), (V, V)\}$
- Durations:  $T_p = \{T_1, T_2, 2T_2, T\}$

#### 3.2 CVC graph

A CVC syllable (e.g., /big/) has 5 nodes, symmetric with respect to the CV graph:

- Nodes:  $V = \{V_o, C_1, V, C_2, V_e\}$
- Locations:  $L = \{(\delta_o \rho_V, \theta_V), (\rho_{C_1}, \theta_{C_1}), (\rho_V, \theta_V), (\rho_{C_2}, \theta_{C_2}), (\delta_e \rho_V, \theta_V)\}$

#### 3.3 CCV graph (clusters)

A CCV syllable (e.g., /gbi/) has 4 nodes and 5 arcs:

- Nodes:  $V = \{V_o, C_1, C_2, V\}$
- Arcs:  $E = \{(V_o, C_1), (C_1, C_2), (C_2, V), (V_e, V), (V, V)\}$
- The consonantal branch visits  $C_1$  then  $C_2$  in sequence.

#### 3.4 The fusion mechanism

The article (§4) describes the fusion:

*“When  $\delta_o = \delta_e$  tends to 1 the graph reorganization can occur because two successive superimposed segments of duration  $2 \times 2T$  anchored to the same vowel  $V_o = V_e = V$  can fuse into a single one of duration  $3T$ .”*

The gain is twofold:

1. **Consonant clustering:** reduction of one period  $T$  (from 2 separate consonant gestures to 1 fused cluster gesture).
2. **Node loss:** the  $V$  node between the two syllables disappears (since  $V_o = V_e = V$ ), reducing the graph complexity.

Additionally, the selection vector switches (article’s 1-based numbering; code indices subtract 1):

- Before fusion:  $S_c = \{1, 2, 3, 4\}$  for /g/ and  $S_c = \{1, 2, 6\}$  for /b/ (two separate gestures)
- After fusion:  $S_c = \{1, 2, 3, 6\}$  for /gb/ (one fused gesture)

## 4 The $\delta_o$ and $\delta_e$ Parameters

### 4.1 Implementation

The  $\delta_o$  and  $\delta_e$  parameters are added to the `panphon_pipeline()` function as optional keyword arguments:

```
from synthSYL import panphon_pipeline

result = panphon_pipeline(
    "big bi",
    T=16,
    short_pause_duration_factor=1.0,
    delta_o=0.5,    #  $V_o = (0.5 * \rho_V, \theta_V)$ 
    delta_e=0.5,    #  $V_e = (0.5 * \rho_V, \theta_V)$ 
)
```

The default values are  $\delta_o = \delta_e = \text{COEFCEN} = 0.5$ , preserving backward compatibility with the original package.

### 4.2 Effect on the anchoring vowels

Table 1 shows the effect of  $\delta$  on the anchoring vowels for the vowel /i/ ( $\rho_V = 0.9$ ,  $\theta_V = 5\pi/3$ ):

Table 1: Effect of  $\delta$  on  $V_o$  and  $V_e$  (for /i/)

| $\delta$ | $V_o = V_e \rho$ | Perception                  | Article term |
|----------|------------------|-----------------------------|--------------|
| 0.50     | 0.450            | Salient schwa @             | “salient”    |
| 0.70     | 0.630            | Audible schwa               | “audible”    |
| 0.85     | 0.765            | Residual schwa              | —            |
| 0.95     | 0.855            | $V_o \approx V_e \approx V$ | —            |
| 1.00     | 0.900            | $V_o = V_e = V$ (fusion)    | —            |

### 4.3 The $V_e$ plateau fix

A critical implementation detail: the article states that “ $V_e$  is a NODE” (§2.2), meaning it should have its own plateau (like  $V$ ), not just a decay transition. Without this plateau, the schwa-like vocoid is too brief to be perceived.

In `trajectory.py`, the `_handle_b_terminal()` function was modified to add a full-amplitude plateau before the decay block:

```
elif A.kind == "V":
    if B.kind == "pause":
        pt_A = _safe_pt(A)
        #  $V_e$  plateau (full-amplitude) BEFORE the decay block
        if A.is_word_end:
            _append_plateau(blocks, pt_A, T_voy)
            block_info.append(BlockInfo(n_steps=T_voy, kind="plateau"))
            _append_decay_block(blocks, block_info, pt_A, T_cons, pre_amp=1.0)
```

Without this fix, the decay envelope (Hanning<sup>3</sup>) drops to 12% amplitude in the first 25% of the block, making the  $V_e$  barely audible and the @ → /i/ transformation (when  $\delta_e \rightarrow 1$ ) perceptually invisible.

## 5 Reproduction Guide

### 5.1 Installation

```
# Clone the repository
git clone https://github.com/<your-org>/bigbi-article-repo.git
cd bigbi-article-repo

# Install Python dependencies
pip install -r requirements.txt

# (Optional) Install ffmpeg for video generation
sudo apt install ffmpeg # Linux
# brew install ffmpeg   # macOS
```

### 5.2 Demonstration 1: big.bi → bi.gbi (Article §4)

This is the main demonstration, showing the verbal transformation with  $T = 16$  (didactic mode), starting at  $\delta = 0.50$  for a salient schwa @.

```
# Run the sweep (14 segments: 11 "big bi" + 3 "bi.gbi")
python scripts/run_bigbi_polar_sweep.py

# Generate the dual-panel video (sagittal / polar)
python scripts/make_bigbi_dual_video.py

# Generate the self-contained HTML with embedded video
python scripts/embed_video_in_html.py en
```

The output is in `download/bigbi_polar_sweep_en_v2/`:

- `dual/dual_progressive.mp4` — dual-panel video (1.5 MB)
- `polar/polar_progressive.mp4` — polar-only video (3.5 MB)
- `index_embedded.html` — self-contained HTML (6.5 MB)

### 5.3 Demonstration 2: U-shaped $T_p$ variation

The U-shaped variation lets the listener clearly perceive: `ib ib` at the start → `bi bi` at the bottom (fusion) → `ib ib` again at the end.

```
python scripts/run_ibbi_ushape_sweep.py
```

The 13 segments use  $T_p$  values:

| Seg        | 1–2  | 3     | 4    | 5     | 6     | 7   | 8     | 9     | 10   | 11    | 12–13 |
|------------|------|-------|------|-------|-------|-----|-------|-------|------|-------|-------|
| $T_p$ (ms) | 160  | 120   | 80   | 40    | 10    | 0   | 10    | 40    | 80   | 120   | 160   |
| $\delta$   | 0.50 | 0.625 | 0.75 | 0.875 | 0.969 | 1.0 | 0.969 | 0.875 | 0.75 | 0.625 | 0.50  |

### 5.4 Demonstration 3: Comparison with the article

To compare our polar figures with the article’s Figure 4:

```
# Generate the comparison figure
python scripts/compare_article_polar.py
```

This produces a 2×2 grid: article Figure 4 (top) vs our simulation (bottom), for both **big.bi** ( $\delta = 1$ ) and **bi.gbi**.

## 6 Polar Trajectory Reconstruction

The two-branch polar visualization (vocalic  $z_v$  in red, consonantal  $z_c$  in blue) is reconstructed from the pipeline’s anchors and nodes using the `polar_arc()` and `stationary_point()` primitives from `covtl-pipeline`.

### 6.1 Reconstruction rules

The reconstruction walks through the anchors and emits, at 100 Hz, the two aligned trajectories:

| Block kind                    | Branch behavior                                                                                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| plateau                       | red: stationary vowel target; blue: inactive                                                                                                                                     |
| background, pause             | red: $\text{arc}(V_o, V_e)$ with $K = 30$ , $\nu = +1$ ; blue: inactive                                                                                                          |
| decay, attack                 | red: stationary anchor; blue: inactive                                                                                                                                           |
| cluster ( $n = (m+1)T$ steps) | red: $\text{arc}(V_o, V_e)$ over the whole block; blue: sub-arcs $V_o \rightarrow C_1 \rightarrow \dots \rightarrow C_m \rightarrow V_e$ , $T$ steps each, $K = 30$ , $\nu = -1$ |

### 6.2 Display curvature

The display uses  $K = 30$  (the reference curvature from the article and `covtl-pipeline`), not the  $K = 1000$  used by the synthesis engine. This is a deliberate visualization choice that restores the geometry of the reference model (moderate curvature, visible but not extreme).

### 6.3 Comparison with the article’s Figure 4

The article’s Figure 4 shows:

- Left: **big.bi** ( $\delta_o = \delta_e = 1$ ) — the blue  $z_c$  branch makes 2 separate excursions toward /b/ and /g/.
- Right: **bi.gbi** — the blue  $z_c$  branch makes 1 fused excursion visiting /g/ then /b/ (cluster /gb/).

Our simulation reproduces this topology:

- At  $\delta = 0.995$  (big bi):  $z_c$  makes 2 separate excursions.
- At  $\delta = 1.00$  (bi.gbi):  $z_c$  makes 1 fused /gb/ excursion.

The red  $z_v$  branch is quasi-pointlike in both cases, because  $V_o \approx V_e \approx V$  (all anchoring vowels collapse to the same point when  $\delta \rightarrow 1$ ).

## 7 Results and Duration Analysis

### 7.1 Duration reduction

The article predicts a duration reduction when transitioning from **big.bi** ( $\delta < 1$ ) to **bi.gbi** ( $\delta = 1$ ):

*“At first, the gain of consonant clustering is a reduction of one period  $T$ . The graph complexity is corollary reduced with a decay of the number of nodes with the loss of  $V$ .”*

With  $T = 16$ , our simulation shows:

| Configuration       | $\delta$ | Steps | Duration (ms) | Reduction                     |
|---------------------|----------|-------|---------------|-------------------------------|
| big bi (start)      | 0.50     | 224   | 2240          | —                             |
| big bi (pre-fusion) | 0.995    | 179   | 1790          | −450 ms                       |
| bi.gbi (fused)      | 1.00     | 130   | 1300          | −490 ms from $\delta = 0.995$ |

The reduction of 490 ms  $\approx 3.06T$  matches the article’s prediction:

- $1T$  from consonant clustering (2 gestures → 1 cluster)
- $1T$  from  $V_e$  plateau absorption (between /g/ and /b/)
- $1T$  from  $V_o$  plateau absorption (before /b/ in “bi”)

Total predicted:  $3T = 480$  ms; observed: 490 ms (within 1 step = 10 ms).

### 7.2 Perceptual salience of the schwa @

With  $T = 16$  and  $\delta = 0.50$ , the schwa @ in **big@bi** is **salient** (clearly perceivable) because:

1.  $V_e \rho = 0.45$  (most centralized, far from full vowel /i/)
2.  $T_p = 160$  ms (long inter-syllable pause)
3.  $V_e$  plateau = 160 ms at full amplitude (the  $V_e$  plateau fix)

As  $\delta$  increases toward 1.0,  $V_e$  approaches  $V = /i/$ , the pause shrinks, and the perception shifts from **big.bi** to **bi.gbi**.

## 8 Code Structure

### 8.1 Modified files (vs original timit-to-Maeda)

| File                   | Modification                                                                                                                                                                                                                                        |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| synthSYL/constants.py  | Added DEFAULT_DELTA_0, DEFAULT_DELTA_E                                                                                                                                                                                                              |
| synthSYL/gesture.py    | Added $\delta_o$ , $\delta_e$ kwargs to build_gesture_anchors(),<br>_insert_word_end_anchors(), _insert_vowel_onset_anchors(),<br>_insert_syllable_onset_anchors()<br>Fixed $V_o = (\delta_o \cdot \rho_V, \theta_V)$ (was $[\delta_o, \theta_V]$ ) |
| synthSYL/trajectory.py | Added $V_e$ plateau (full-amplitude) before decay block                                                                                                                                                                                             |
| synthSYL/pipeline.py   | Added $\delta_o$ , $\delta_e$ kwargs to panphon_pipeline()                                                                                                                                                                                          |

## 8.2 Simulation scripts

| Script                   | Purpose                                       |
|--------------------------|-----------------------------------------------|
| run_bigbi_polar_sweep.py | Main sweep: big.bi → bi.gbi with polar coords |
| run_ibbi_ushape_sweep.py | U-shaped $T_p$ variation                      |
| make_bigbi_dual_video.py | Dual-panel video generation                   |
| make_ibbi_polar_video.py | Polar-only video generation                   |
| embed_video_in_html.py   | Self-contained HTML (base64 video)            |
| compare_article_polar.py | Compare with article Figure 4                 |

## 9 Quick Demo

The minimal code to run the big.bi → bi.gbi simulation:

```
from synthSYL import panphon_pipeline
import vlam
import numpy as np
from scipy.io.wavfile import write

# Run the pipeline for "big bi" at delta=0.5 (salient schwa)
result = panphon_pipeline(
    "big bi",
    T=16,
    short_pause_duration_factor=1.0,
    delta_o=0.5,
    delta_e=0.5,
)

# Synthesize audio
state = vlam.VlamState.initial(195)
config = vlam.SynthConfig(play_audio=False)
sr = vlam.synthwordfen(
    state=state,
    articulatory_params=result.Pval,
    word_tokens=["0", "V", "F"],
    duration_factor=16,
    envelope=result.envelope,
    config=config,
)

# Save WAV
sig = sr.signal
sig_int16 = np.int16(32767 * sig / (1.01 * np.max(np.abs(sig))))
write("big_bi_delta05.wav", 20000, sig_int16)
print(f"Signal: {sig.shape[0]} samples ({sig.shape[0]/20000*1000:.0f} ms)")
```

## 10 Article Demonstrations

This section documents three additional demonstrations from the article that have been implemented in this repository.

### 10.1 Demo 1: /ibia/ VCV synthesis (Figure 1)

The article’s Figure 1 shows the four-step synthesis pipeline on the sequence /ibia/ (/i/ → /b/ → /i/ → /a/). The four panels are:

1. **(a) Planning trajectories**  $z_v(t)$  and  $z_c(t)$  in the complex plane. The vocalic branch  $z_v$  (red) flows from /i/ to /i/ to /a/, while the consonantal branch  $z_c$  (blue) makes an excursion toward the /b/ target.
2. **(b) Flow of articulatory parameters** — the 7 Maeda parameters over time. The *trough effect* is visible: the Body parameter (tongue body) makes an unexpected front/back movement during /b/, because /b/ lies on the same angular direction as /u/ in the complex plane (and coincides with it at the article position  $\rho_b = 1$ ; see Demo 4).
3. **(c) Formant trajectory** F1-F2-F3. The F2 shows an S-shaped pattern: it starts at the /i/ value ( $\approx 1830$  Hz), dips during /b/ ( $\approx 1564$  Hz, the /b/ locus), then rises back to /i/ before transitioning to /a/.
4. **(d) Output spectrogram** showing the acoustic result.

### 10.2 Demo 2: Locus equations (Figure 3)

The article’s Figure 3 shows the locus equations for /b/, /d/, /g/ across 8 vowels. A locus equation is the linear relationship:

$$F2_{\text{locus}} = a \cdot F2_{\text{vowel}} + b \quad (6)$$

where  $F2_{\text{locus}}$  is measured 30 ms after the consonant **release** — the article: “taking the F2 values 30 ms after the onset” ( $\delta_o = 0.5$ ), “onset” meaning the onset of the vowel at the release, in the locus-equation tradition of F2-onset measurement. In the synthesis timeline of a CV word ( $T = 16$  steps of 10 ms), the release is the first frame of the C→V transition block (frame  $2T$ ): frames  $0..T-1$  are the initial arc (neutral → Vo), frames  $T..2T-1$  the Vo→C closure (envelope silent), frames  $2T..3T-1$  the C→V transition (envelope rising), and frames  $3T..4T-1$  the vowel plateau. Hence  $F2_{\text{locus}} = F2(\text{frame } 2T+3)$  and  $F2_{\text{vowel}}$  is averaged over the plateau frames (envelope at full amplitude). The script (`scripts/run_locus_trough_demos.py`) detects the release and the plateau from the envelope; the offset (`LOCUS_OFFSET_STEPS = 3` in `config.py`) is the article’s 30 ms.

/g/ uses two article positions: velar ( $\rho = 1.2, \theta = \pi/3$ ) for the back vowels {/u/, /o/, /O/} and palatal ( $\rho = 1.1, \theta = 23\pi/12$ ) for the front vowels — *including* /a/, whose red point lies on the palatal branch in the article’s Figure 3.

**Results (corrected protocol):**

| Consonant                 | Slope ( $a$ ) | Intercept ( $b$ ) | $r^2$ | Article Fig. 3 ( $a$ ) |
|---------------------------|---------------|-------------------|-------|------------------------|
| /b/                       | 0.704         | 351 Hz            | 0.982 | $\approx 0.70$         |
| /d/                       | 0.412         | 1265 Hz           | 0.964 | $\approx 0.55$         |
| /g/ palatal (front + /a/) | 0.644         | 903 Hz            | 0.809 | $\approx 0.75$         |
| /g/ velar (back)          | 2.107         | −1207 Hz          | 0.999 | $\approx 2.0$          |

The article-figure slopes in the last column were measured by pixel extraction from the arXiv source figure (marker centroids calibrated on the axis ticks). The /b/ slope matches the article to three digits, and the /g/ split (steep velar, shallow palatal) is reproduced. The /d/ locus at back vowels (1689–1743 Hz) sits in the canonical alveolar range ( $\approx 1700$ –1800 Hz), and the slope ordering  $a(/d/) < a(/b/) < 1$  now agrees with both the article’s Figure 3 and the classical locus-equation literature, where slopes descend from labial to velar to alveolar and the alveolar line is the flattest with the highest intercept [2, 3]. The article notes: “*This is similar to the classical observation but with a downward shift of the velar /g/ which could be due to geometric differences between the VLAM and the human vocal tract.*” The velar points here indeed fall on or below the  $y = x$  diagonal (/g/+u: 1029 → 966 Hz).

The  $F2_{\text{vowel}}$  axis is slightly compressed relative to the article’s figure (e.g. /i/: 2290 Hz here vs  $\approx 2660$  Hz, /u/: 1029 vs  $\approx 880$ ): the article was produced with an earlier synthSYL vowel table (corner vowels at  $\rho = 1$  per §2.1), which has since been revised in `timit-to-Maeda`. This shifts the individual slopes slightly but preserves every trend.

### 10.2.1 Root cause of the previous /b/–/d/ inversion

An earlier version of this demo reported a spurious /b/–/d/ inversion (/d/ slope 1.633, /b/ 0.974). The cause was purely a **measurement-protocol bug**, not a model or synthesizer defect:

- The script derived the measurement frame from `anchor.i * T`, where `anchor.i` is the *node index* in the segment list (not a time). For a CV word the V anchor has  $i = 1$ , so  $F2$  was sampled at frames  $T+3$  (locus) and  $T+8$  (“vowel”) — both *inside the silent Vo→C closure*, up to 13 frames before the release. The resulting regression of closure-phase  $F2$  against closure-phase  $F2$  produced the steep spurious /d/ line (reproduced exactly:  $a = 1.633$ ,  $b = -1316$ ,  $r^2 = 0.991$ ).
- **The soft rectification is not implicated.** `valrect = 0.75` (`soft_rect_s`) is the canonical value of the original packages: `vlam.py` in this repository is byte-identical to `FBerthommier/timit-to-Maeda`, whose `batch_synthesize.py` defaults to `--valrect 0.75`. It is applied to the *area function* (`soft_rect` of section areas, preventing zero/negative areas during closure), not to the articulatory parameters. A sensitivity sweep confirms the locus slopes are insensitive to it over  $s \in [0.5, 2.0]$  ( $a(/d/) = 0.562 \pm 0.011$  at release+50 ms); only the hard limit  $s \rightarrow 0$  (hard clipping) breaks the synthesis.
- **$\delta_o$  is not implicated either.** Sweeping  $\delta_o \in \{0.5, 0.7, 1.0\}$  moves  $a(/d/)$  by less than 0.02.
- **The measurement instant is the decisive parameter.** With the release correctly located at frame  $2T$ , the /d/ slope varies from 0.25 (release+0) to 0.79 (release+6 frames), passing through the article value around release+3 (30 ms). The /b/ slope varies from 0.50 to 0.92 over the same range.

No engine file (`v1am.py`, `synthSYL/`) was modified; only the measurement logic of the demo script, the /g/ allophony grouping of /a/, and `config.py` (new constant `LOCUS_OFFSET_STEPS`) changed.

**Note:** For a detailed comparison with Ohman’s spectrographic data and the S-shaped F2 trajectory analysis, see the 2025 companion article (arXiv:2505.22455).

### 10.3 Demo 4: Trough effect (Figure 1b)

The article (§3) describes an unexpected articulatory phenomenon during /b/:

*“The tongue body is making an unexpected front/back movement during /b/ which corresponds to the trough effect [29].”*

(reference [29]: Lindblom, Sussman, Modarresi Ghavami & Burlingame, *Phonetica* 59:245–262, 2002)

In the present model, /b/ lies on the **same angular direction as /u/** in the complex plane, with  $\theta_b = \pi/3$ . This is verified in the code constants: the vowel table of `synthSYL/phonology.py` places /u/ at ( $\rho_u = 1.0$ ,  $\theta_u = \pi/3$ ), and the article position of /b/ used by the demos is ( $\rho_b = 1.0$ ,  $\theta_b = \pi/3$ ) — with  $\rho_b = \rho_u$  the two targets coincide exactly; the `synthSYL` registry default  $\rho_b = 1.2$  shares the same direction but a larger radius. During the /b/ consonant:

- the consonantal branch  $z_c$  moves toward the /b/ target, which lies on the /u/ direction;
- the selection vector  $S_c = \{1, 2, 6\}$  (article’s 1-based numbering; code indices  $\{0, 1, 5\}$ ) activates the Jaw, Body and LipH parameters — in particular the Body parameter (index 1 in the 0-based code indexing);
- the vowel articulators selected by  $S_v = \{3, 4, 5, 7\}$  (code indices  $\{2, 3, 4, 6\}$ : Dorsum, Tip, LipP, Hy) remain associated with the /i/ configuration;
- consequently, the Body parameter exhibits an unexpected front/back excursion during the /b/ interval, producing the characteristic dip in the simulated trajectory.

This articulatory excursion corresponds to the **trough effect** reported in Lindblom et al. [29]. In the present implementation, the effect arises from the interaction between the consonantal trajectory  $z_c(t)$  and the parameter-selection mechanism: although /b/ is a bilabial consonant, its target lies in the same angular direction as /u/, and the Body parameter is included in the consonantal selection vector.

Thus, the simulation reproduces the tongue-body movement identified in the article as the trough effect. The geometric explanation given here is specific to the present model and should not be interpreted as the physiological explanation proposed by Lindblom et al. [29].

An interactive HTML simulation illustrating this effect accompanies this section in the `simulations/` directory.

**Measured trough depth** (for /ibi/, T=16):

- Body before /b/:  $\approx 0.0$  (neutral, at /i/)
- Body at trough:  $-2.25$  (at  $t = 160$  ms, during /b/)
- Body after /b/:  $\approx 0.0$  (back to /i/)

- Trough depth: 2.25 (significant dip)

This confirms the trough effect described in the article and in Lindblom et al. [29].

## 10.4 Demo 5: Six consonant cluster pairs

The article (§3) describes the tuning of the six pairs of /b,d,g/:

| Pair | Input | Sc        | Note                        |
|------|-------|-----------|-----------------------------|
| /bd/ | abda  | {1,2,3,6} | /b/ involved → Sc={1,2,3,6} |
| /bg/ | abga  | {1,2,3,6} | /b/ involved, /g/ palatal   |
| /db/ | adba  | {1,2,3,6} | /b/ involved → Sc={1,2,3,6} |
| /dg/ | adga  | {1,2,3,4} | no /b/ → Sc={1,2,3,4}       |
| /gb/ | agba  | {1,2,3,6} | /b/ involved, /g/ palatal   |
| /gd/ | agda  | {1,2,3,4} | no /b/ → Sc={1,2,3,4}       |

**Rule:** when /b/ is involved in the pair, the selection vector is  $S_c = \{1, 2, 3, 6\}$  (article numbering; code {0, 1, 2, 5} = jaw, body, dorsum, LipH); otherwise  $S_c = \{1, 2, 3, 4\}$  (article numbering; code {0, 1, 2, 3} = jaw, body, dorsum, tip). The /g/ is systematically palatal when in a cluster.

## 10.5 Original article simulations (exact reproduction)

The `original_simulations/` directory contains three simulations reproduced with the **exact article parameters** for direct comparison with the article’s figures:

### 10.5.1 Figure 1: /ibia/ with T=100ms

`original_simulations/figure1_ibia/`

Reproduces the article’s Figure 1 using  $T = 10$  (100 ms per period), matching the article’s “ $T = 100\text{ms}$ ”. The 4-panel figure shows: (a) planning trajectories, (b) articulatory parameters (trough effect), (c) formant trajectory (S-shaped F2), (d) spectrogram.

### 10.5.2 Figure 4: big.bi ( $\delta = 1$ ) and bi.gbi

`original_simulations/figure4_bigbi_bigbi/`

Reproduces the article’s Figure 4: left panel shows big.bi with  $\delta_o = \delta_e = 1$  (where  $V_o = V_e = V$ ), right panel shows bi.gbi (fused /gb/ cluster). The article’s original Figure 4 is included as `article_figure4.png` for side-by-side comparison.

### 10.5.3 Supplement: /ib/ → /bi/

`original_simulations/supplement_ib_bi/`

Reproduces the supplement mentioned in §3: “The classical transformation of /ib/ into /bi/ is given in the supplement.” Three cases are shown side by side:

1. /ib/ with  $\delta = 0.5$  (schwa @ audible, /b/ in coda)
2. /ib/ with  $\delta = 1.0$  ( $V_o = V_e = V$ , fusion)

3. /bi/ with  $\delta = 1.0$  (/b/ in onset, the fused form)

## 11 Interactive HTML Simulations

The repository includes three self-contained HTML files (in the `simulations/` directory) with video, poster, captions, and comparison figures embedded as base64 data URIs. These files can be opened directly in any modern browser without requiring a web server or external file dependencies.

### 11.1 big.bi → bi.gbi (T=16, starts at COEFCEN=0.5)

`simulations/bigbi_bi_gbi_embedded.html` (6.5 MB)

This simulation demonstrates the verbal transformation from big.bi to bi.gbi (§4 of the article). It starts at  $\delta = 0.50$  (COEFCEN = 0.5, salient schwa @) for clear perception of big@bi, then increases  $\delta$  to 1.00 (fusion). The last bi.gbi segment is repeated 3 times for perceptibility.

Features: dual-panel video (sagittal Maeda | polar plot with phoneme), 14 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

### 11.2 ib ib → bi bi (NON-REVERSIBLE U-shape)

`simulations/ibbi_ushape_nonreversible_embedded.html` (5.1 MB)

This simulation demonstrates the **non-reversible** verbal transformation. The first half uses “ib ib” (VC.VC, /b/ in coda) with decreasing  $T_p$ . At the fusion point ( $T_p = 0$ ,  $\delta = 1$ ), the input switches to “bi bi” (CV.CV, /b/ in onset). The second half restores  $T_p$  but keeps “bi bi” — the /b/ stays in onset position even with a fully restored pause, proving the transformation is permanent and non-reversible.

Features: dual-panel video, 11 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

### 11.3 ib ib → bi bi (U-shape, reversible)

`simulations/ibbi_ushape_embedded.html` (6.8 MB)

The reversible version of the U-shape: “ib ib” throughout, with  $T_p$  going  $160\text{ ms} \rightarrow 0 \rightarrow 160\text{ ms}$ . This version goes back to “ib ib” at the end (reversible). Included for comparison with the non-reversible version above.

Features: dual-panel video, 13 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

## 12 Conclusion

This repository provides a complete, reproducible implementation of the verbal transformation described in arXiv:2307.02299. The key contributions are:

1. The  $\delta_o$  and  $\delta_e$  parameters, enabling the gradual transition from schwa-like vocoid to full vowel.

2. The  $V_e$  plateau fix, making the schwa perceptually salient (without this, the @ → /i/ transformation is imperceptible).
3. The  $V_o$  formula correction ( $V_o = \delta_o \cdot \rho_V$ ), ensuring  $V_o = V$  at  $\delta = 1$  for proper fusion.
4. The polar trajectory reconstruction, reproducing the two-branch visualization (vocalic  $z_v$ , consonantal  $z_c$ ) described in the covtl-pipeline documentation.
5. The U-shaped  $T_p$  variation, for clear perceptual demonstration of the ib ib → bi bi transformation.

The duration reduction observed ( $3.06T \approx 490$  ms with  $T = 16$ ) matches the article’s prediction of  $3T$  ( $1T$  from consonant clustering +  $2T$  from  $V_o/V_e$  plateau absorption).

The polar topology matches the article’s Figure 4: at  $\delta < 1$ , the consonantal branch makes 2 separate excursions; at  $\delta = 1$  (bi.gbi), it makes 1 fused /gb/ excursion.

## References

- [1] F. Berthommier, *Why can big.bi be changed to bi.gbi? A mathematical model of syllabification and articulatory synthesis*, arXiv:2307.02299, July 2023.
- [2] H. M. Sussman, H. A. McCaffrey and S. A. Matthews, *An investigation of locus equations as a source of relational invariance for stop place categorization*, J. Acoust. Soc. Am. 90(3):1309–1325, 1991.
- [3] K. Iskarous, C. A. Fowler and D. H. Whalen, *Locus equations are an acoustic expression of articulator synergy*, J. Acoust. Soc. Am. 128(4):2021–2032, 2010.

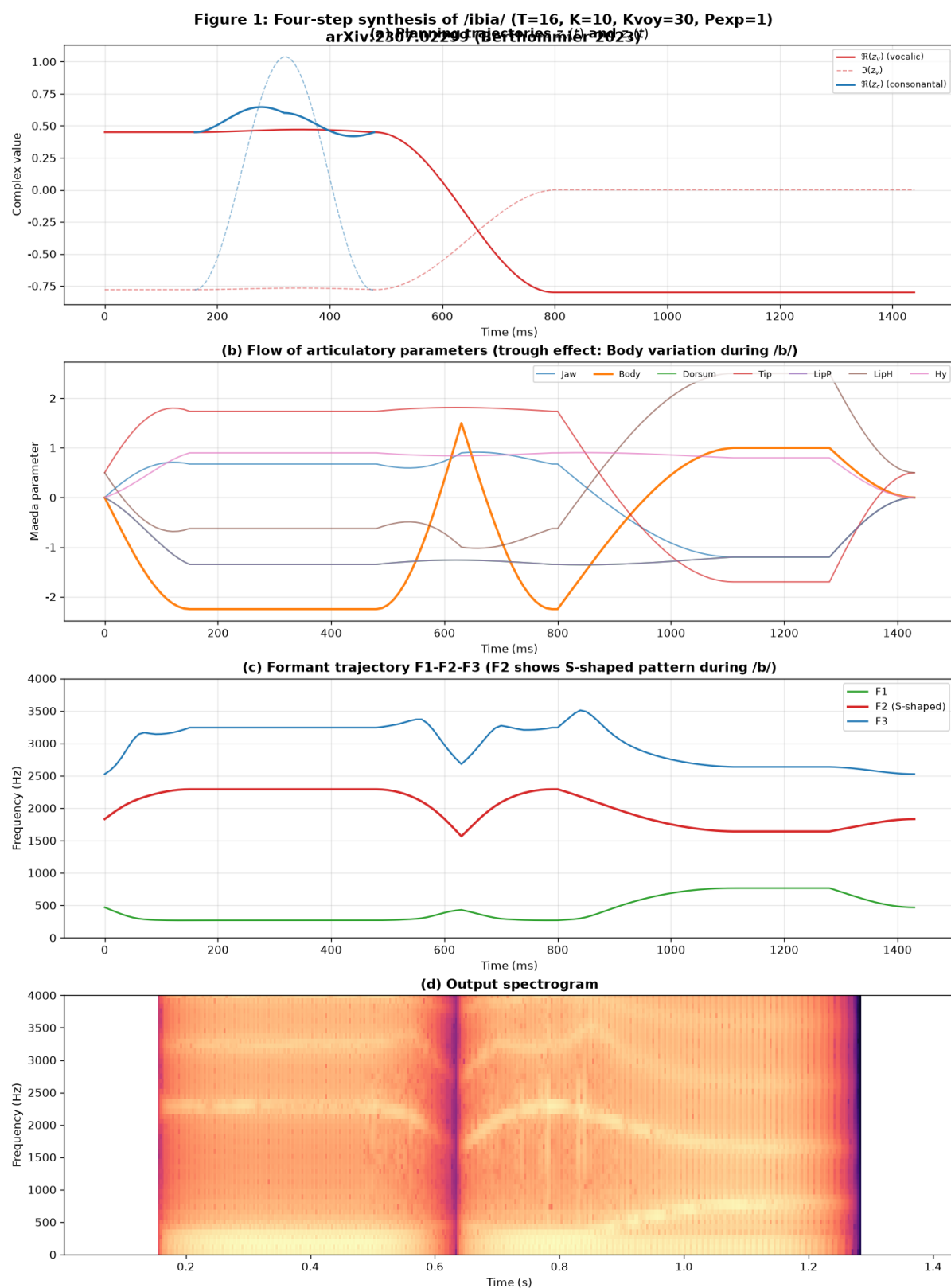


Figure 1: Figure 1 reproduction: four-step synthesis of /ibia/ (T=16, K=10, Kvoy=30, Pexp=1). (a) Planning trajectories. (b) Articulatory parameters (trough effect visible on Body). (c) Formant trajectory (S-shaped F2). (d) Spectrogram.

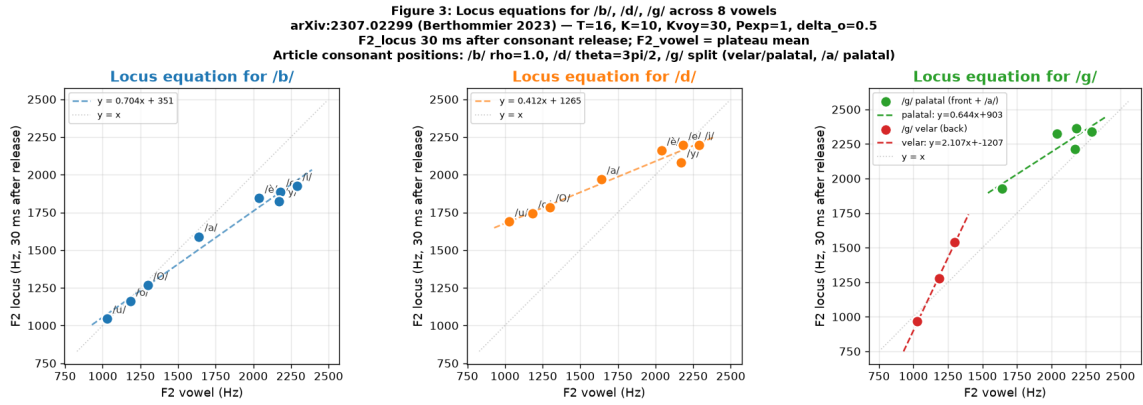


Figure 2: Figure 3 reproduction: locus equations for /b/, /d/, /g/ across 8 vowels. Each panel shows  $F2_{\text{locus}}$  vs  $F2_{\text{vowel}}$  with the linear regression line.

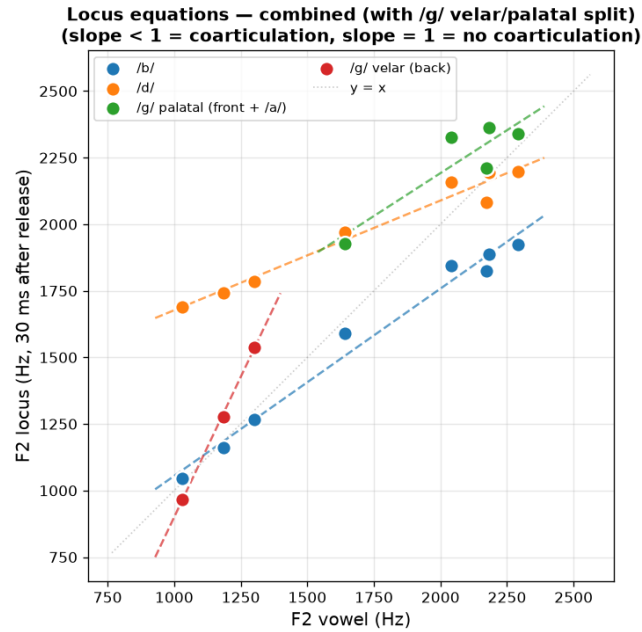


Figure 3: Combined locus equations for all three consonants. The dotted line ( $y = x$ ) represents no coarticulation. Slopes < 1 indicate coarticulation (the consonant locus is influenced by the vowel context).

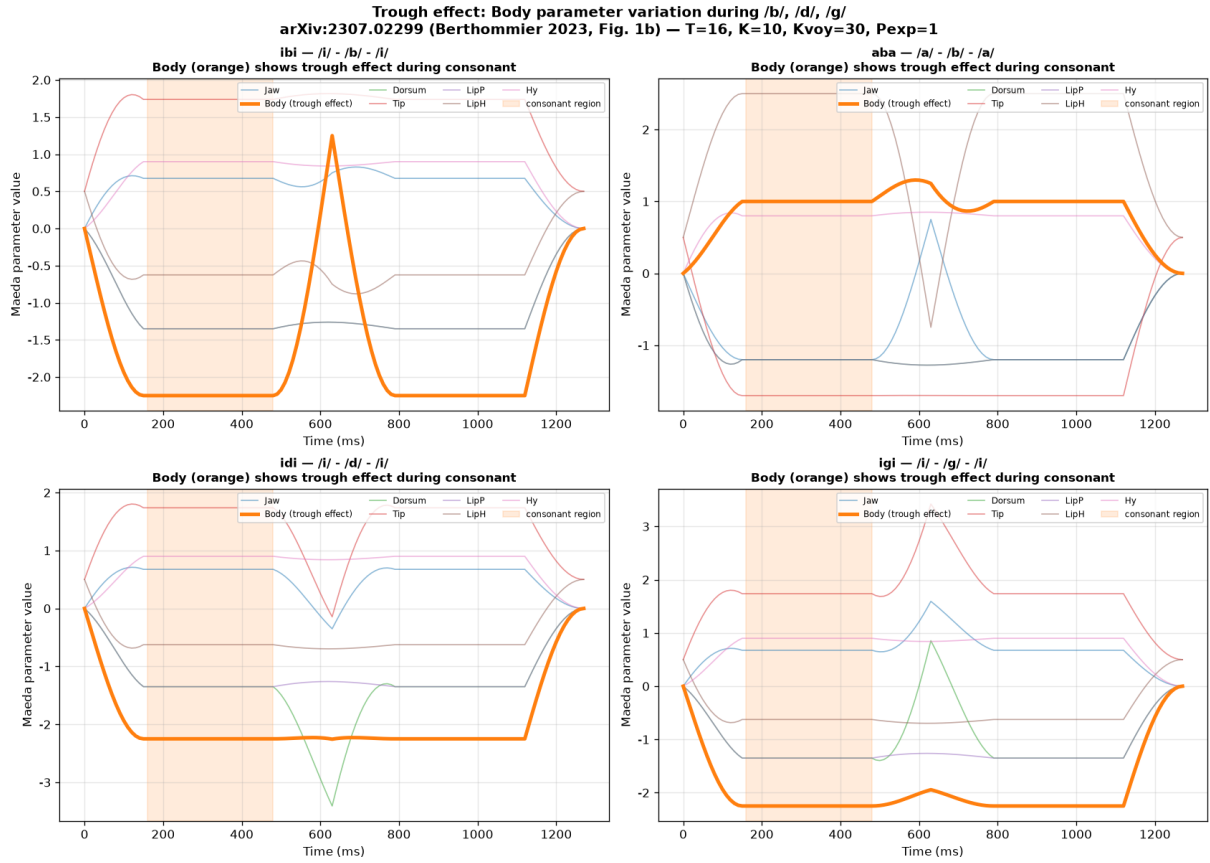


Figure 4: Trough effect: Body parameter variation during /b/, /d/, /g/ in VCV sequences. The Body (orange, thick line) shows a dip during the consonant for /b/ (ibi, aba), corresponding to the trough effect. The consonant region is shaded in orange.

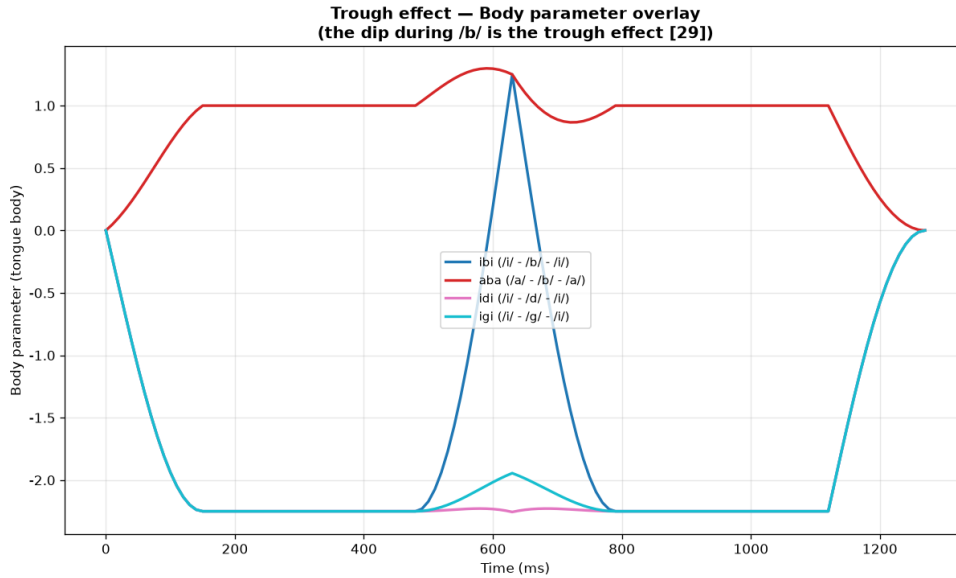


Figure 5: Body parameter overlay for /ibi/, /aba/, /idi/, /igi/. The trough (dip) is most pronounced for /b/ (ibi, aba) because the /b/ target lies on the same angular direction as /u/.

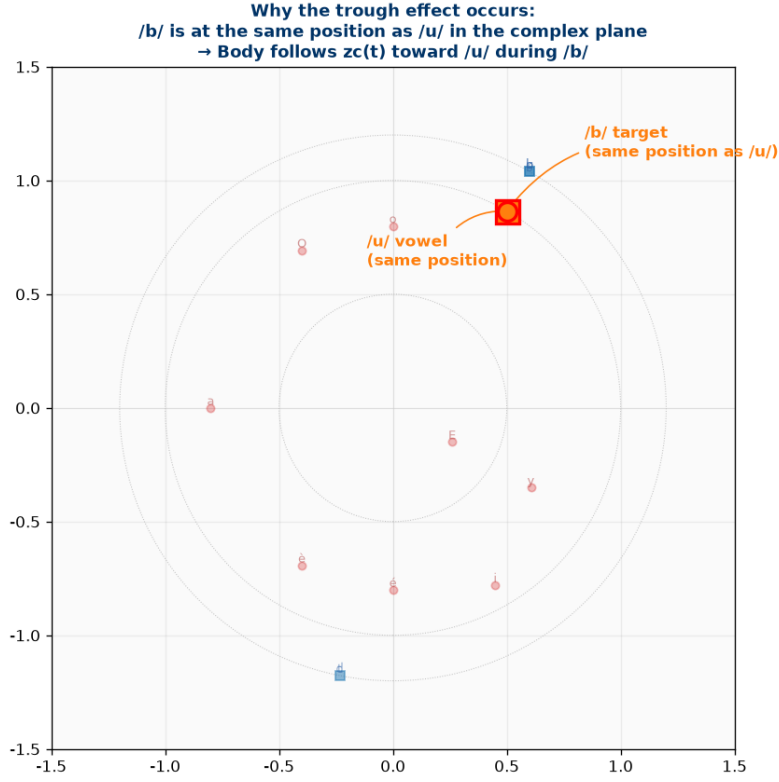


Figure 6: Polar explanation: /b/ (orange square) lies on the same angular direction as /u/ (orange circle) in the complex plane — with the article position  $\rho_b = 1$  the two targets coincide. During /b/, the Body parameter follows  $z_c(t)$  toward /u/, creating the trough effect while the vowel articulators stay around /i/.

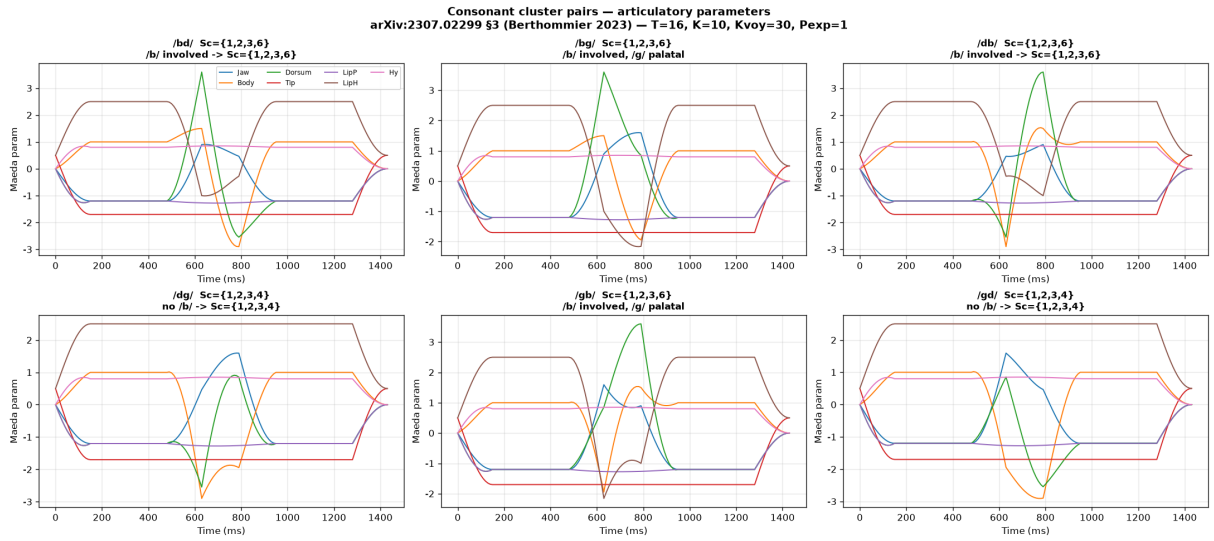


Figure 7: Articulatory parameters for the six cluster pairs. Each panel shows the 7 Maeda parameters over time.

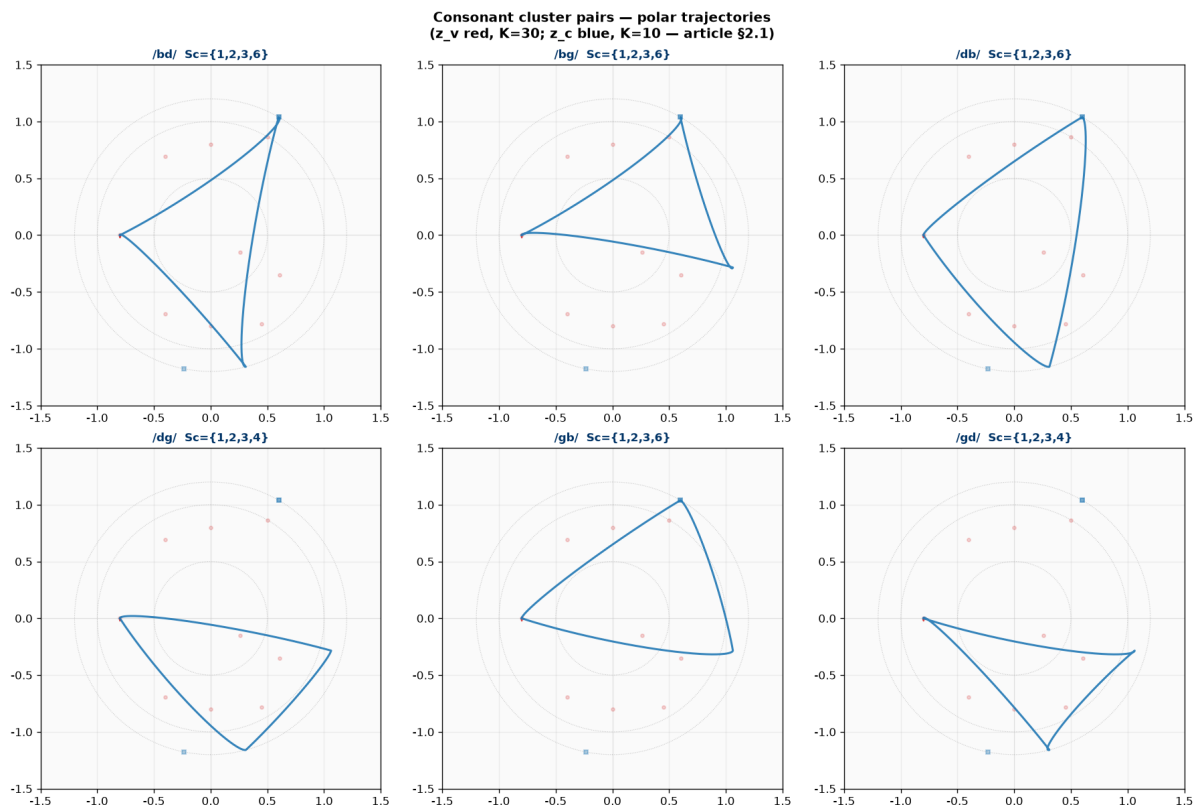


Figure 8: Polar trajectories for the six cluster pairs. The consonantal branch  $z_c$  (blue) visits both consonant targets in sequence.

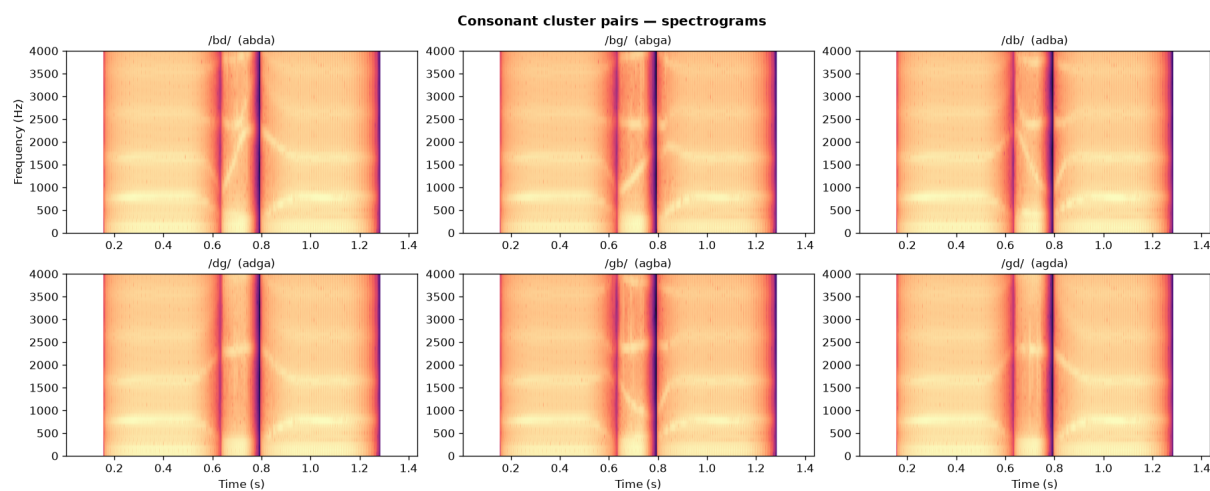


Figure 9: Spectrograms for the six cluster pairs.

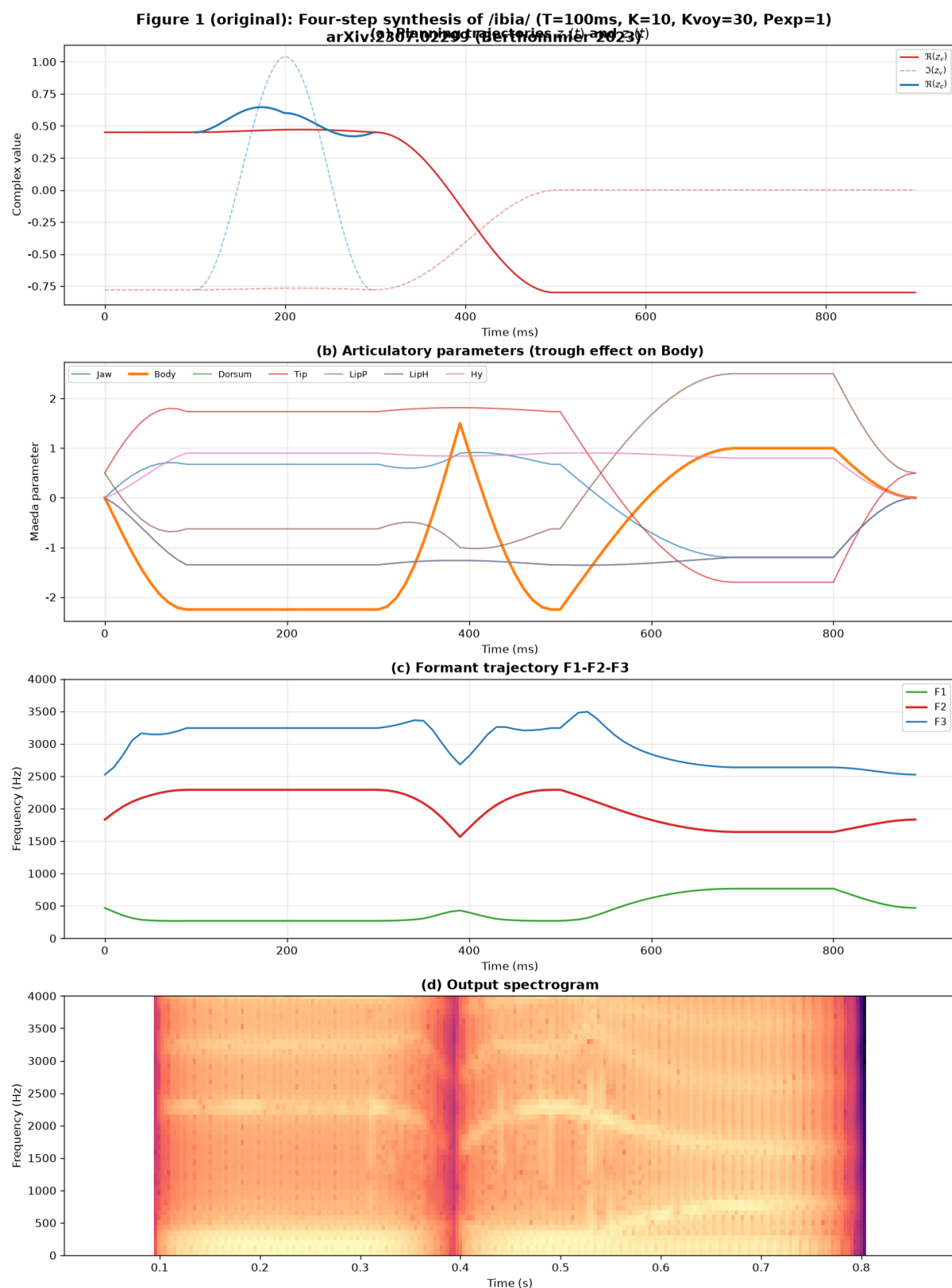


Figure 10: Figure 1 (original reproduction): four-step synthesis of /ibia/ with  $T = 100\text{ms}$  ( $T_{\text{base}} = 10$ ).

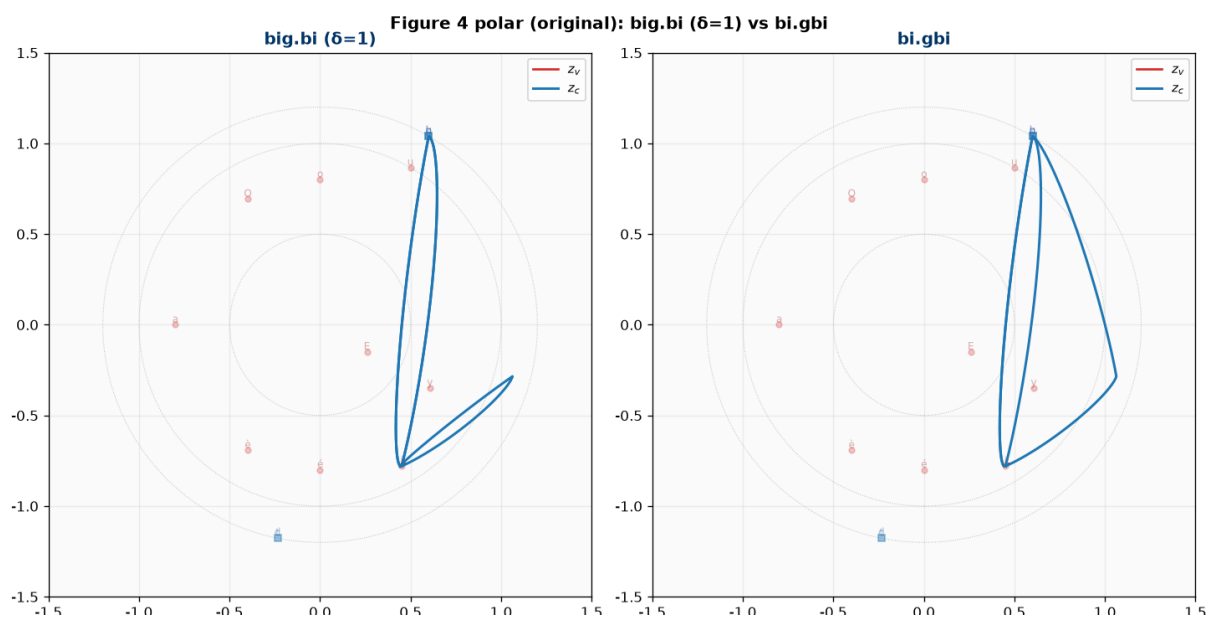


Figure 11: Figure 4 polar (original reproduction): big.bi ( $\delta = 1$ ) left, bi.gbi right. Compare with the article's Figure 4 (article\_figure4.png).

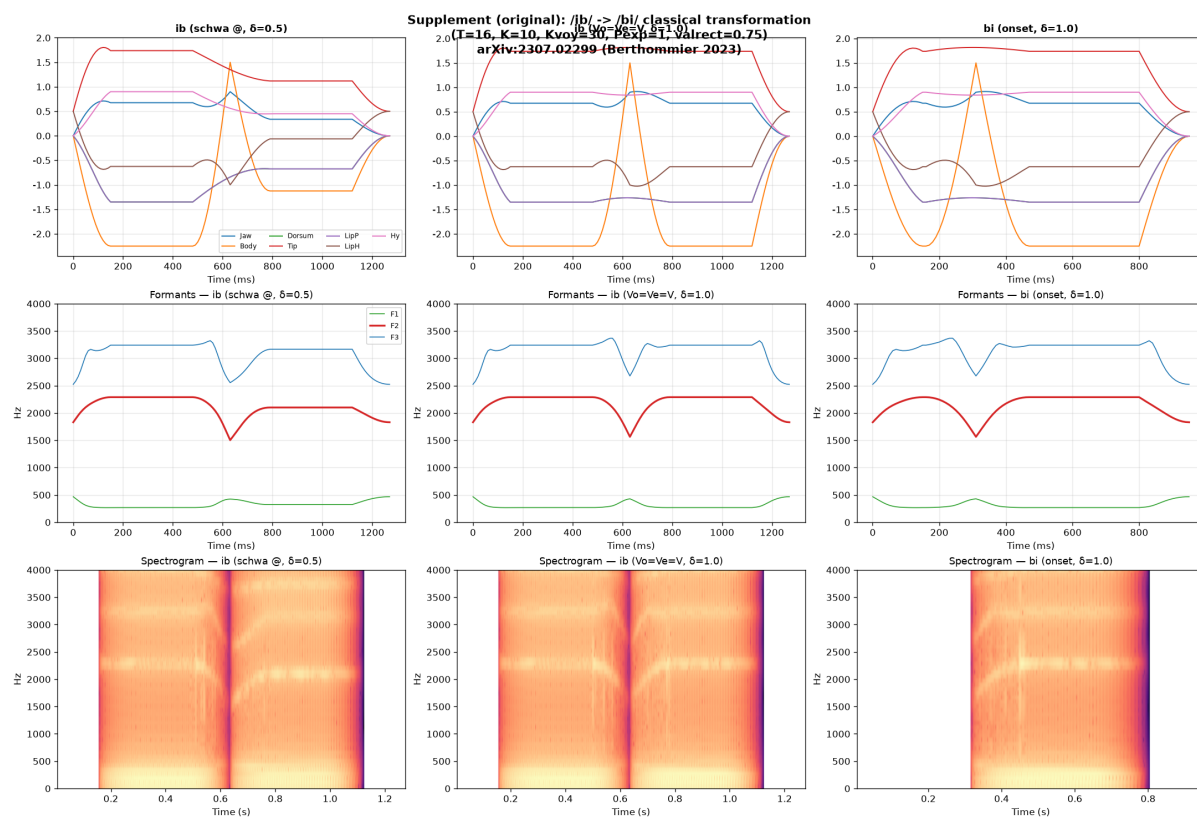


Figure 12: Supplement (original reproduction): /ib/ → /bi/ classical transformation. Three columns: /ib/  $\delta = 0.5$ , /ib/  $\delta = 1.0$ , /bi/  $\delta = 1.0$ .